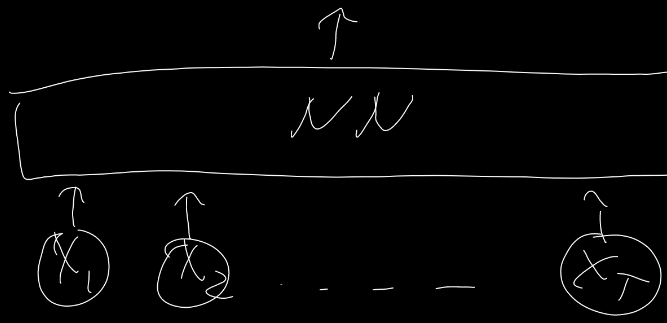
NNs for texts



Stemming

modelling
models
modelled

$\rightarrow$ model

Lemmatization

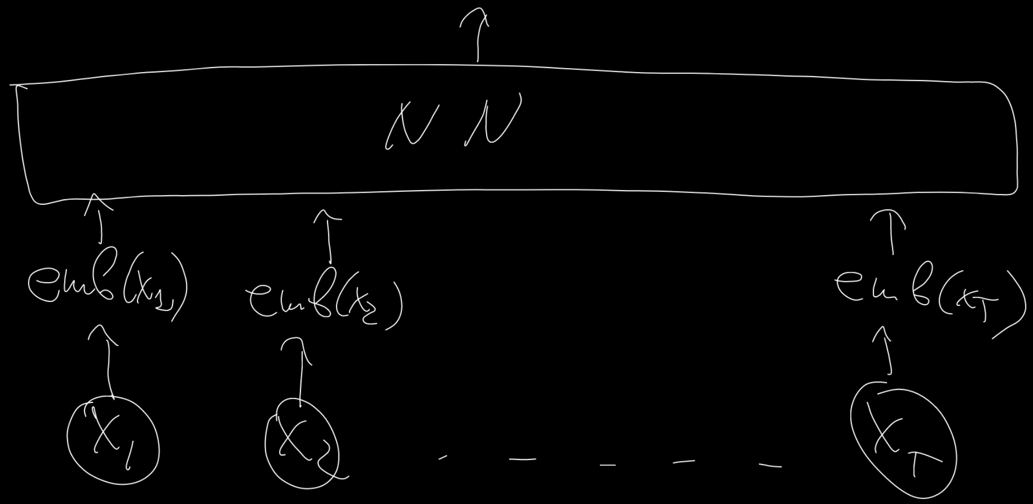
went $\rightarrow$ go

lem(modelling, pos="verb") = model
lem(modelling, pos="subject") = modelling

stop words: "the", "a", "in", "after"

pipeline: texts → preprocessing → words → remove stop words → stemming lemmatization → remove rare words

preprocessing:
to lower case
remove URLs
remove commas, ..



$x_t \in \{1, 2, \dots, V\} \to$

$\to \mathrm{emb}(x_t) \in \mathbb{R}^{\mathrm{emb\_dim}}$

Byte Pair Encoder (BPE)

- "hug" – 10 (h, ug; h, ug)
- "pug" – 10
- "pun" – 12
- "bun" – 4
- "hugs" – 5

Vocab: h, u, g, p, n, b, s

All pairs: hu – 15
hg – 0
ug – 25

Next all pairs: hug – 15
hu – 16

add

add

inference "hugs" → greedy tok → hug, s

↳ apply merging rules in order → hu, gs

Vocab: "hu", "gs", "hug"

WordPiece tokenizer

word → w, ##o, ##r, ##d

w, ##o → wo

##o, ##r → ##or

$$\text{score}(\text{token1}, \text{token2}) = \frac{\text{freq}(\text{token1} \,\&\, \text{token2})}{\text{freq}(\text{token1})\,\text{freq}(\text{token2})}$$

un, ##able

bre, ##men → bremen

Unigram tokenizer

hug $- 10 - p_1$

pug $- 5 - p_2$

pun $- 12 - p_3$

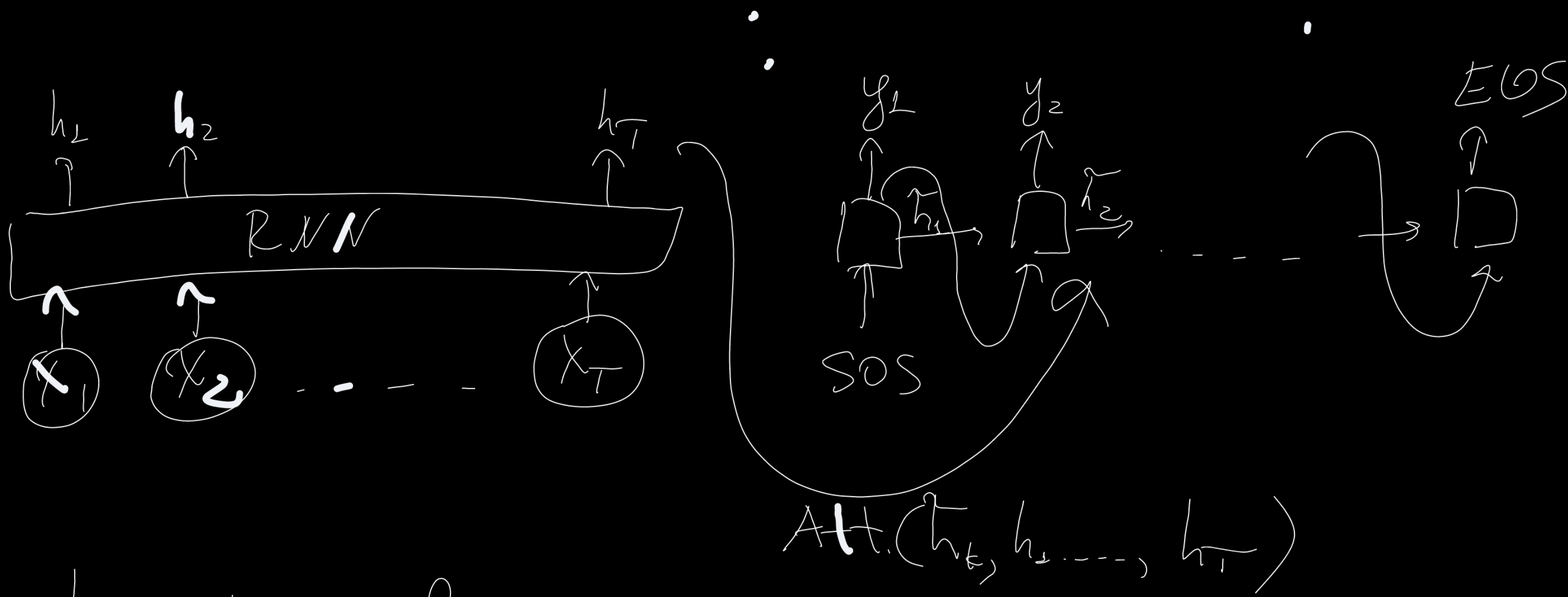
bun $- 4 - p_4$

hugs $- 5 - p_5$

h, u, g, p, n, b, s, hu, ug, pu, un, bu, gs, hug, ugs

hug → $p(\text{"hu"}, \text{"g"}) = p(\text{"hu"})\,p(\text{"g"}) = \frac{15}{210} \cdot \frac{20}{210}$

$\text{loss} = -10 \log p_1 - 5 \log p_2 \ldots$



teacher forcing

problem: exposure bias

with p: take token from training

$1-p$: take token from model prediction

Beam search

$$\log p(y_1 \ldots y_t \mid x, \theta) + \alpha \log p_{LM}(y_1 \ldots y_t \mid x, \theta) + \beta t \to \max$$

language model

$$p(y \mid x, \theta) = p(y_1 \mid x, \theta)\, p(y_2 \mid y_1, x, \theta) \ldots p(y_M \mid y_1 \ldots y_{M-1}, x, \theta)$$

- $$\arg\max_y p(y \mid x, \theta) = \arg\max_y \prod_{t=1}^{M} p(y_t \mid y_1 \ldots y_{t-1}, x, \theta)$$

V^M elements

$t=1$		$t=2$		$t=3$
a – 0.5	top-2	aa – 0.5 · 0.5		aaa
b – 0.4		ab – 0.5 · 0.4		aab
c – 0.1		ac – 0.5 · 0.1		aac
		ba – 0.4 · 0.45		aba
		bb – 0.4 · 0.45		abb
		bc – 0.4 · 0.45		abc
		ca – 0.1 · 0.5		
		cb – 0.1 · 0.4		
		cc – 0.1 · 0.1		

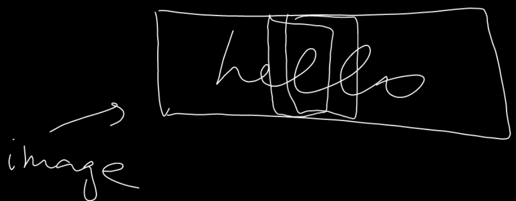
top-2 →

$O(MKV)$

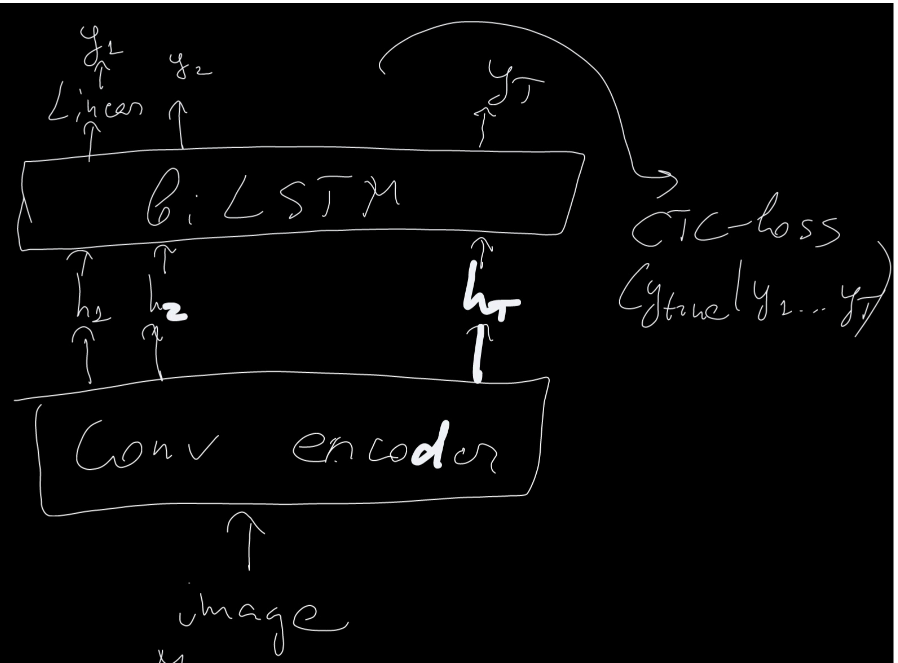
Only on inference!

CTC-loss

handwritten text recognition



→ h, e, l, l, o

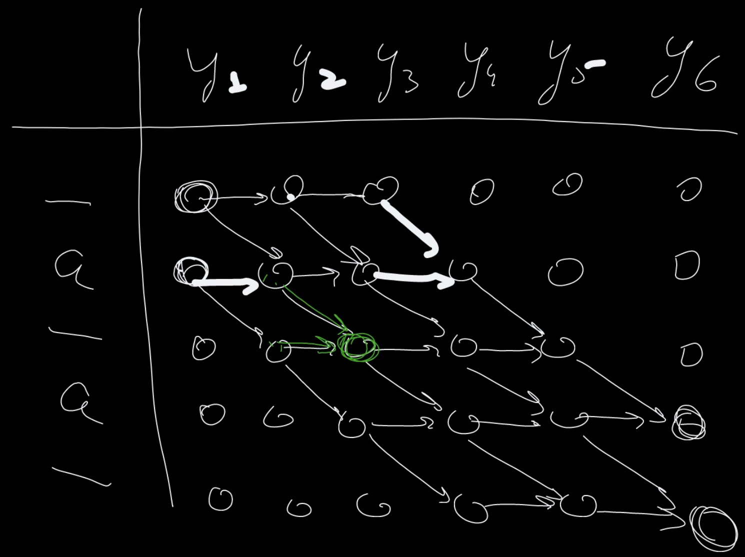


$$p_{CTC}(y_{true} \mid y_1 \dots y_M, x, \theta) = \sum_{y:\, CTC(y) = y_{true}} \prod_{t=1}^{M} p_t(y_t \mid x, \theta)$$

"hello"

CTC(_ _ h _ ee _ l _ l _ o) → _ h _ e _ l _ l _ o → hello

Example: y_{true} = "aa", $M = 6$



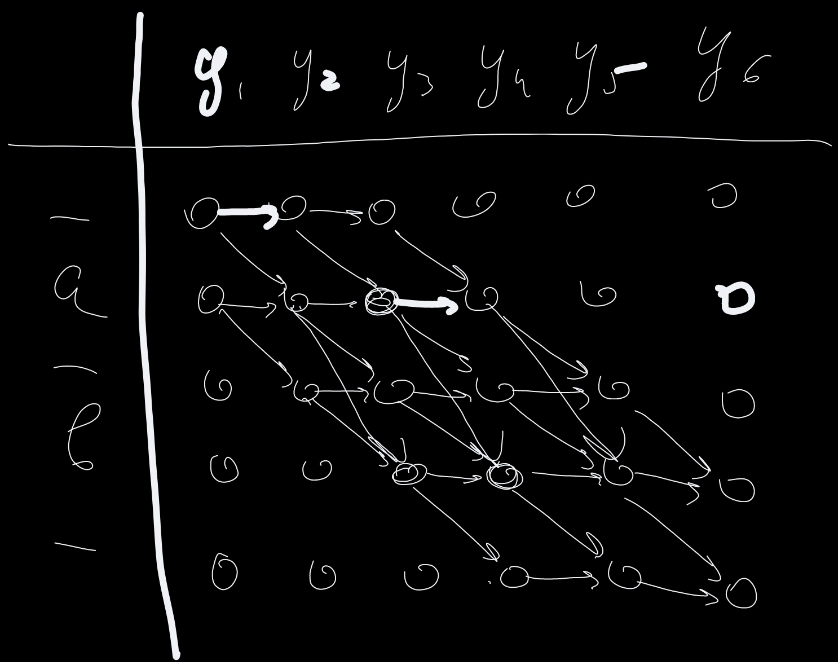
$$CTC(\_\ \_\ \_\ a\ \_\ a) = aa$$

$$\alpha_{s,t} = p_t(y_t \mid x, \theta)\left(\alpha_{s,t-1} + \alpha_{s-1,t-1}\right)$$

$$CTC = \alpha_{4,6} + \alpha_{5,6}$$

$$O(M \cdot len(y_{true}))$$

$y_{true} = \text{"ab"}, \; M = 6$



$$\alpha_{s,t} = p_t(y_t \mid x, \theta)\left(\alpha_{s,t-1} + \alpha_{s-1,t-1} + \alpha_{s-2,t-1}\right)$$

if $y_{true,s} \neq \_$ and

$y_{true,s} \neq y_{true,s-2}$

$$\alpha_{s,t} = p_t(y_t \mid x, \theta)\left(\alpha_{s,t-1} + \alpha_{s-1,t-1}\right)$$

otherwise

$$CTC = \alpha_{4,6} + \alpha_{5,6}$$

Non-differential losses

$BLEU(y_{true}, y)$ – machine translation

$CIDEr(y_{true}, y)$ –

image-to-caption

a b a c

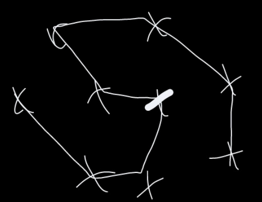
1-gramms a, b, c

2-grams, ab, ba, ac

3-gram, aba, bac

TSP

$$\sum_i dist(y_i, y_{i+1})$$



non-diff. loss

$$F(\theta) = \mathbb{E}_{p(y|x,\theta)} f(y) \to \min_\theta$$

Problem:
very high variance!

$$\nabla_\theta F(\theta) = \nabla_\theta \mathbb{E}_{p(y|x,\theta)} f(y) = \nabla_\theta \sum_y p(y|x,\theta) f(y) =$$

$$= \sum_y \nabla_\theta p(y|x,\theta) f(y) = \sum_y p(y|x,\theta) \frac{\nabla_\theta p(y|x,\theta)}{p(y|x,\theta)} f(y) =$$

$$= \mathbb{E}_{p(y|x,\theta)} \nabla_\theta \log p(y|x,\theta) f(y) \approx \qquad \nabla_\theta \log p(y|x,\theta)$$

$$\approx \frac{1}{M} \sum_{j=1}^{M} \nabla_\theta \log p(y_j|x,\theta) \left(f(y_j) - b(x) \right), \quad y_j \overset{i.i.d.}{\sim} p(y|x,\theta)$$

put this to SGD, ADAM

REINFORCE

log-derivative trick

Baseline $b(x)$

$$\nabla_\theta F(\theta) = \mathbb{E}_{p(y|x,\theta)} \nabla_\theta \log p(y|x,\theta)\left(f(y) - b(x)\right)$$

$$\mathbb{E}_{p(y|x,\theta)} \nabla \log p(y|x,\theta)\, b(x) = b(x) \sum_y p(y|x,\theta) \nabla \log p(y|x,\theta) =$$

$$= b(x) \sum_y \cancel{p(y|x,\theta)} \frac{\nabla_\theta p(y|x,\theta)}{\cancel{p(y|x,\theta)}} = b(x) \nabla_\theta \underbrace{\sum_y p(y|x,\theta)}_{=1} = 0$$

Good $b(x)$: 1 $\mathbb{E}_{p(y|x,\theta)} f(y)$

2 $f(y_{greedy\_rollout})$